

# Oluwatobi Melvyn Mayungbo

ML Engineer, MLOps and Production Systems

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## SUMMARY

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ML engineer focused on reproducible experimentation. Built a pipeline with MLflow tracking, rolling-origin backtesting and leakage tests, benchmarked deep sequence models across seeds against strong baselines, and served a model at 5.1 ms p99 with ONNX Runtime. Writes clean, tested Python.

## SKILLS

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**Engineering and MLOps:** Python, MLflow, Prefect, Docker, Pytest, Git, SQL, DuckDB, PostgreSQL, Redis, FastAPI, GCP Cloud Run, AWS ECS Fargate, Railway, dbt-core, Evidently, CI/CD (GitHub Actions), Streamlit, React

**Machine Learning:** PyTorch, LightGBM, Scikit-learn, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

**Statistics and Causal Inference:** Time-Series Backtesting, Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Logistic Regression, WOE Scorecards

**AI and LLM Systems:** ONNX Runtime, Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face

**Data Analysis:** Pandas, NumPy, SQL, Anomaly Detection

**Payments and Risk:** Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

## PROJECTS

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### Self-Operating Demand Forecasting Platform | [GitHub](#)

*Tech: Python, Prefect, dbt-core, DuckDB, LightGBM, MLflow, Evidently, FastAPI, Docker, GitHub Actions*

- Built an end-to-end hourly demand forecasting system for NYC taxi zones, producing 24-hour quantile forecasts through an orchestrated Prefect pipeline spanning ingestion, dbt/DuckDB transformations, backtesting, drift monitoring and conditional retraining.
- Validated with rolling-origin backtesting across 6 expanding-window folds against seasonal-naive and historical-mean baselines, reporting MASE and pinball loss, with automated leakage tests that fail the build if any feature reads past the forecast origin.

### Sequence-Based Transaction Fraud Detection System | [GitHub](#)

*Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP*

- Built and compared three deep sequence architectures (GRU, Temporal Convolutional Network and a causally masked Transformer encoder) over customer transaction histories, with learned embeddings for high-cardinality merchant (~2.9k vocabulary), MCC and device-type categoricals; all models under 115k parameters and CPU-trainable in under 6 minutes.
- Benchmarked all architectures against a strong LightGBM baseline on engineered velocity and deviation features, reporting PR-AUC and precision at fixed review capacity (top 0.1%, 0.5%, 1%) across 3 seeds, with a per-attack-type breakdown covering card testing, account takeover and merchant compromise; sequence models cut expected cost per transaction 27% under an explicit cost model.
- Deployed real-time scoring by exporting to ONNX Runtime (2.9× faster than eager PyTorch at batch size 1) and serving via FastAPI, reaching 5.1 ms p99 end-to-end latency against a 50 ms budget, with append-only per-decision audit logging and attention-based attribution of the prior transactions driving each flag.

### Premier League Match Predictor | [GitHub](#)

*Tech: Python, LightGBM, scikit-learn, pandas, SQLite, Flask, Vercel*

- Backtested gameweek by gameweek over 1,050 matches: 52.1% outcome accuracy against 43.2% for always picking the home side, and log loss 0.9948 against 1.0061 for an Elo-only baseline, with well calibrated probabilities.

### Production-Grade Double-Entry Ledger API | [GitHub](#)

*Tech: Python 3.12, FastAPI, PostgreSQL 16, Redis 7, Alembic, Docker*

- Load tested to find throughput saturating at about 250 to 260 req/s; removing the hot account cut transfer p99 from 1800 ms to 390 ms (4.6×) at 50 users.

## EXPERIENCE

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**Machine Learning Instructor, SQI College of ICT** March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

**Internal Auditor, NISER** May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

## EDUCATION

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**B.Sc. Statistics, University of Ibadan** January 2021 to March 2025

**Professional Diploma in Artificial Intelligence, SQI College of ICT** July 2025 to Present

## CERTIFICATIONS

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- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action